# Cardiac Assessment for Patients Undergoing Noncardiac Surgery

## A Multifactorial Clinical Risk Index

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**• In this article, we describe a multifactorial cardiac risk index that can be used to assess patients undergoing noncardiac surgery. The index is a modified version of an index that was previously generated by Goldman and coworkers on a set of 1001 consecutive patients and prospectively validated in our clinical setting (a general medical consultation service in a large teaching hospital) on 455 patients. We present a Bayesian approach to assessing cardiac risks by converting average risks for patients undergoing particular surgical procedures (pretest probabilities) to average risks for patients with each index score (posttest probabilities). A simple nomogram is presented for performing such a calculation.**
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Cardiologists and general internists are frequently asked to perform cardiac risk assessments for patients undergoing noncardiac surgery. Prior to 1977, most studies of preoperative cardiac risk factors examined the relationship between individual variables, such as recent myocardial infarction,[1] and the risk of perioperative cardiac events. In 1977, Goldman et al[2,3] published the first multivariate approach studying 1001 consecutive patients undergoing noncardiac surgery at the Massachusetts General Hospital, Boston. A nine-variable index was created by exploring their data set using step-wise linear discriminant analysis. The index was shown to have good predictive properties in the "derivation (test) data set" (ie, in the data from which the index was generated).

We have recently tested the predictive validity of a modified version of the original multifactorial index in a prospective fashion. We modified the index in two ways: we incorporated additional variables that we believed to be clinically important at the outset of the study, and we simplified the scoring scheme. We then applied this index to 455 consecutive patients undergoing noncardiac surgery who were referred to our general medical consultation service. The occurrence of perioperative cardiac events was determined by an individual who was blinded to the preoperative score. We have reported the study design, details of statistical analysis, and outcome of the study elsewhere.[4] In this article, we describe our clinical use of the index in detail and make recommendations for its use in other clinical settings.

### DESCRIPTION OF INDEX

Table 1 shows the variables and weighting scheme for our modified multifactorial index. The variables are divided into seven groups.

The first group concerns symptoms and history of coronary artery disease as follows: a myocardial infarction that has occurred within six months of the planned surgery; a myocardial infarction that has occurred more than six months before the planned surgery; Canadian Cardiovascular Society (CCS) angina class 3 or 4[5] in the two weeks prior to the planned surgery; and unstable angina that has occurred within three months of the planned surgery. Class 3 angina is defined as angina that occurs when the patient walks one to two blocks on a level surface and climbs one flight of stairs under normal conditions and at a normal pace. Class 4 angina is defined as an inability to carry on any physical activity without discomfort. Angina may or may not be present at rest. Unstable angina is defined as follows: crescendo angina (more severe, prolonged, or frequent angina superimposed on an existing pattern of relatively stable exertion-related angina); coronary insufficiency syndrome; new onset angina (within one month) brought on by minimal exertion; and angina brought on at rest, as well as with minimal exertion. We do not consider variant, or Prinzmetal's, angina as unstable angina, although some cardiologists do. That is, if a patient has a pattern of angina that is mostly atypical (ie, occurs principally at rest and not with exertion), we do not increase his index score by the ten points assigned to the variable *unstable angina* as listed in Table 1.

A history of alveolar pulmonary edema results in the addition of either ten or five points to the index score, depending on its timing as shown in Table 1. For recent pulmonary alveolar edema, we require chest roentgenographic documentation. If the patient has a remote episode, we would accept a convincing clinical history in the absence of a chest roentgenogram. (The patient often tells us that he entered another hospital in severe respiratory

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distress and was treated with diuretics and was subsequently told that he had "water on the lung" or congestive heart failure as a cause of this respiratory distress.)

Twenty points are added to the index score if the patient is suspected of having critical aortic stenosis. This assessment is made on the basis of classic features in the history (near syncope, exertional angina, or recurrent congestive heart failure) in the setting of other signs (pulsus parvus et tardus, a thrusting left ventricular impulse in the presence of a low blood pressure, and left ventricular hypertrophy). We recognize the difficulty of making a diagnosis of suspected critical aortic stenosis on the basis of signs and symptoms. In particular, the intensity of the murmur may not be predictive of the gradient. Nevertheless, internists and cardiologists are able to make an assessment of this variable, albeit with low reliability.[4] If the clinician would refer the patient for cardiac catheterization because of suspected critical aortic stenosis, then we consider this characteristic to be present in the patient and 20 points are added to the index score. Where available, the gradient across the aortic valve can be assessed by the use of Doppler echocardiography.[6]

There are two kinds of arrhythmias that are included in the index. If the patient has a sinus rhythm with atrial premature beats or a rhythm other than sinus (eg, atrial fibrillation) on the last electrocardiogram obtained prior to the planned surgery, five points are added to the index score. If the patient has a history of more than five premature ventricular contractions at any time prior to surgery, five points are added to the index score. This latter characteristic includes a history of ventricular tachycardia or ventricular fibrillation at any time prior to surgery.

The variable *poor general medical status* is defined in the same fashion as that presented in the original study of Goldman et al.[2] The features are listed in Table 1. If the patient is 70 years old or older, five points are added to the score. A surgical procedure is designated as an *emergency operation* if the patient must be taken to the operating room before his or her medical management can be optimized (eg, fluid status, electrolyte imbalance, adjustment of antianginal regimen, etc).

Table 1.—Modified Multifactorial Index

| Variables | Points |
|---|---|
| Coronary artery disease | |
| Myocardial infarction within 6 mo | 10 |
| Myocardial infarction more than 6 mo | 5 |
| Canadian Cardiovascular Society angina | |
| Class 3 | 10 |
| Class 4 | 20 |
| Unstable angina within 3 mo | 10 |
| Alveolar pulmonary edema | |
| Within 1 week | 10 |
| Ever | 5 |
| Valvular disease | |
| Suspected critical aortic stenosis | 20 |
| Arrhythmias | |
| Sinus plus atrial premature beats or rhythm other than sinus on last preoperative electrocardiogram | 5 |
| More than 5 ventricular premature beats at any time prior to surgery | 5 |
| Poor general medical status* | 5 |
| Age over 70 years | 5 |
| Emergency operation | 10 |

*Oxygen pressure <60 mm Hg; carbon dioxide pressure >50 mm Hg; serum potassium <3.0 mEq/L (<3.0 mmol/L); serum bicarbonate <20 mEq/L (<20 mmol/L); serum urea nitrogen >50 mg/dL (>18 mmol/L); serum creatinine >3 mg/dL (>260 mmol/L); aspartate aminotransferase, abnormal; signs of chronic liver disease; and/or bedridden from noncardiac causes.

## ESTIMATING RISK OF CARDIAC EVENTS FOR INDIVIDUAL PATIENTS

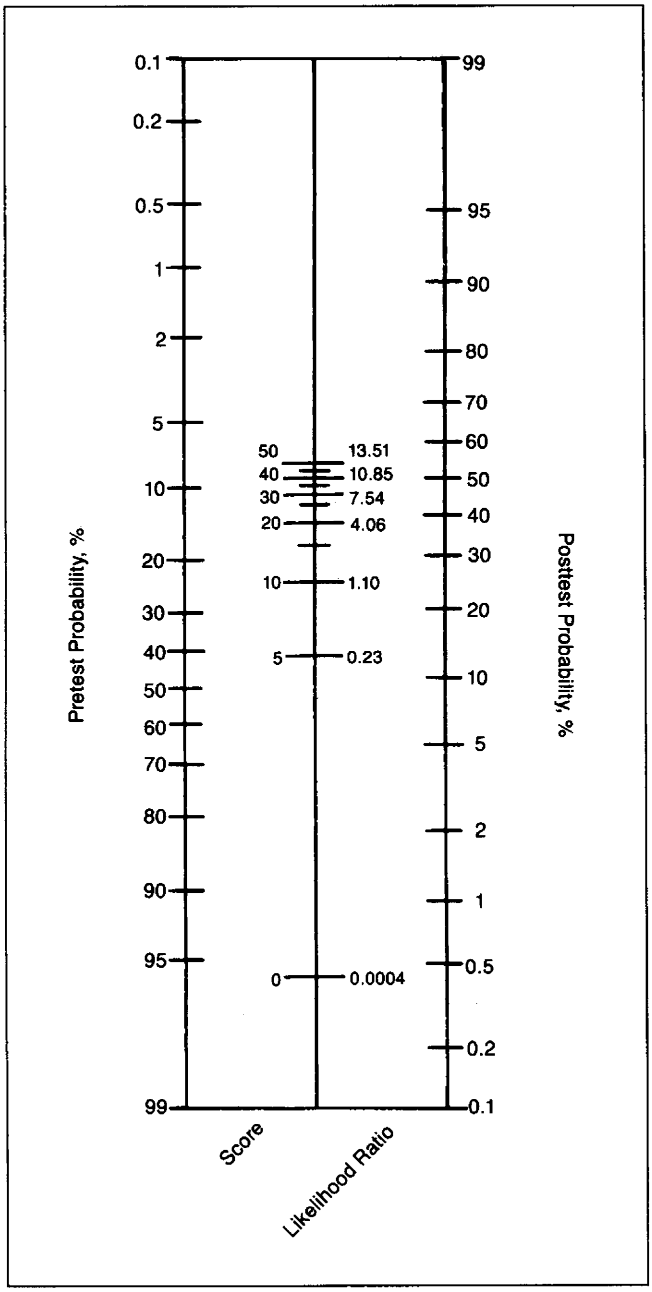
We describe the following four steps for estimating the risk of perioperative cardiac events.

### Step 1: Determine Average Risk for Surgical Procedure in Your Clinical Setting

Even before consideration of an individual patient's cardiac characteristics, those who assess cardiac risk must consider the average risk of a perioperative cardiac complication in the clinically relevant population of all patients undergoing noncardiac surgical procedures (Table 2). This risk will vary depending on the type of surgical procedure, characteristics of the patient population at your institution, and mechanism of referral to your clinical practice. For example, patients undergoing a repair of an abdominal aortic aneurysm will have a higher chance of developing a cardiac complication than those undergoing a cataract extraction. Similarly, patients in a tertiary facility with expertise in cardiac anesthesia may differ from those who have their surgery at community hospitals. Lastly, patients referred to a cardiologist for cardiac risk assessment will probably have a higher average risk than a series of consecutive patients undergoing surgery. Characteristics such as the type of surgery and referral patterns for the institution and clinical practice should be considered in assessing the average risk for such a procedure. This average risk, which is independent of any individual patient's characteristics but depends on the characteristics of all of the patients in the population defined as relevant for your clinical setting, is called the *pretest probability*, where the test is the individual patient's index score.

We calculated these average risks, or pretest probabilities, for patients in our clinical practice undergoing various surgical procedures. The reader should note that our clinical practice is a general medical consultation service at a large tertiary facility that has a wide referral network. Our sample (denominator) included only those patients referred by the surgical services to the general medical consultation service in whom a question of cardiac risk arose either during the admission history and physical examination (performed by the surgical service) or during our preoperative medical consultation obtained for other reasons (eg, diabetes mellitus or chronic lung disease). The cardiology service at our hospital refers virtually all preoperative risk assessments to the general medical consultation service; thus, we feel that these patients are similar to those who would be seen by cardiologists and therefore would have the same pretest probability (average risk). On the other hand, this is not a sample of consecutive patients undergoing surgery, and, therefore, our pretest probabilities will be higher than those that would be found in such a series (eg, the study by Goldman et al[2]).

While our study[4] incorporated many kinds of cardiac complications, ranging from minor events such as postoperative angina to cardiac death, we have found that surgeons and anesthetists are most interested in predicting three kinds of events: cardiac death, nonfatal myocardial infarction, and nonfatal alveolar pulmonary edema. Table 2 shows the average risk of developing these severe cardiac complications (pretest probabilities) for the patients in our series grouped into various surgical procedures. While there were 30 such complications in total in our series, nine of them were fatal. Thus, roughly one third of the pretest probability is accounted for by the risk of cardiac death.




Likelihood ratio nomogram. Anchor straight edge at value on pretest side of nomogram determined by surgical procedure. Direct straight edge through point in center column reflecting patient's index score and associated likelihood ratio. Point where straight edge meets right-hand column denotes patient's posttest probability, ie, his risk of perioperative cardiac complication.

### Step 2: Calculate Risk Score

Using the scoring system shown in Table 1, we calculate a risk score. Although the maximum possible score is 105 points, very few patients have a score greater than 40.

### Step 3: Use a Nomogram to Convert Average Risk for Procedure (Pretest Probability) to Average Risk for Patient (Posttest Probability)

The information about an individual patient contained in the risk index score is used to convert the pretest probability (or average risk of a severe cardiac complication for all patients undergoing a similar surgical procedure) to a posttest probability (average risk for a patient with a similar index score). The technique involves the use of likelihood ratios, which are described in more detail elsewhere.[7] The likelihood ratio for any risk score is derived by dividing the proportion of patients found to have that risk score among those who develop cardiac complications by the proportion of patients who had that same score among those who did not develop cardiac complications. A likelihood ratio of one (eg, a risk score of 10) implies that the pretest and posttest probabilities are equal. That is, the patient's risk is equal to the average risk for patients undergoing that procedure. A likelihood ratio of less than one implies a less than average risk (posttest probability less than pretest probability), while a likelihood ratio of greater than one implies a greater than average risk (posttest probability greater than pretest probability). Thus, for patients with scores of 0 and 5, the estimated risk of developing a perioperative complication will be lower than the average risk for the procedure. For risk scores greater than 10, the estimated risk will be higher than average. The nomogram shown in the Figure allows a simple conversion of pretest to posttest probabilities by converting all risk index scores to the corresponding likelihood ratios in the middle vertical line. The legend for the Figure describes the use of this nomogram.

Table 2.—Pretest Probabilities for Types of Surgery General Medical Consultation Service (Toronto General Hospital)

| | Severe Cardiac Complications, No. (%)* | 95% Confidence Limits |
|---|---|---|
| Major surgery | | |
| Vascular | 10/76 (13.2) | 6.5-19.7 |
| Orthopedic | 9/66 (13.6) | 6.0-21.2 |
| Intrathoracic/intraperitoneal | 7/88 (8.0) | 3.4-13.6 |
| Head and neck | 1/38 (2.6) | 0.3-16.4 |
| Minor surgery (eg, transurethral prostatectomies, cataracts) | 3/187 (1.6) | 0.5-1.6 |

*Cardiac death, nonfatal myocardial infarction, or nonfatal alveolar pulmonary edema.

### Step 4: Adjust for Other Patient Characteristics Not Considered by Index

Occasionally, patients will present with certain characteristics that are either not included in the index we use or are not adequately considered. For example, a risk score associated with a patient who has a left atrial myxoma may be quite low, while the patient may have a higher than average risk of a perioperative cardiac complication. This occurs because the variable *left atrial myxoma* does not appear in the index. (In order to study the risk associated with a left atrial myxoma, an extremely large sample size or unusual sample of patients would have to be studied.) Similarly, this index will not account for the patient we sometimes encounter who has minimal objective criteria on which one would base a high estimate of risk, but nevertheless "appears" to be a poor-risk candidate. The estimates of posttest risk should not be taken literally if one's subjective clinical judgment suggests a revision. The multifactorial index is merely a "model" or starting point for clinicians, who should not hesitate to make such a revision in appropriate circumstances.

## REPORT OF CASES

CASE 1.—A 67-year-old woman was admitted for an elective cholecystectomy. Her cardiac history included a myocardial infarction that occurred one year prior to this admission. This was followed by frequent angina. Over the last month, her angina had increased in frequency and severity and was now provoked by minimal exertion (eg, getting out of bed and walking to the bathroom). She also had some angina at rest. There was no history of congestive heart failure. Physical examination revealed no cardiac murmurs, and her carotid upstroke was normal. Her electrocardiogram revealed normal sinus rhythm with no premature beats. There was no documented history of ventricular ectopic beats in the past. She had none of the features listed under the poor-general-medical-status category. The positive features of the risk score were as follows: myocardial infarction over six months (five points); CCS angina class 4 (20 points); and unstable angina (ten points), for a total score of 35 points. Note that because the patient's angina had increased to class 4 during the period immediately prior to this admission she received both 20 points for CCS class 4 and ten points for the *unstable-angina-within-three-months* variable.

The pretest probability associated with the intraperitoneal procedure is 8% (Table 2). Using the likelihood ratio nomogram in the Figure, this pretest probability is converted (by the risk score of 35) to a posttest probability of 45%. Our impression, therefore, was that this patient had a 45% chance of developing a severe cardiac complication, such as nonfatal myocardial infarction, nonfatal alveolar pulmonary edema, or cardiac death. Roughly one third of this risk (15%) was her estimated chance of dying during the procedure. The possible reversible element of her risk was her angina class. It would therefore be appropriate to consider changing this woman's medical management, if it was felt to be suboptimal at the present time, or refer her for cardiac catheterization as a step toward a possible coronary artery bypass procedure.

CASE 2.—An 80-year-old man was admitted for a transurethral resection of the prostate. Our consultation service was asked to see him because of a history of ventricular ectopic beats. The patient had stable angina, which came on only with severe exertion. There was no history of myocardial infarction, congestive heart failure, or syncope. The patient was noted to have frequent (>5/min) ventricular premature beats (VPBs) on previous electrocardiograms. He had never been treated with any medication for this problem. On his preoperative electrocardiogram, he was again shown to have greater than five VPBs per minute. A cardiogram was otherwise unremarkable, and he was in sinus rhythm. He had none of the features listed in the poor-general-medical-status category. The procedure was not an emergency.

This patient gets a risk score of ten based on the *age-over-70-years* variable (five points) and "more than five VPBs per minute at any time prior to surgery" (five points). The pretest probability of 1.6% is therefore converted to a posttest probability of 1.8%.

Thus, our impression was that the patient had a very low risk of developing a severe cardiac complication and an extremely low risk (approximately 0.6%) of suffering from a cardiac death during the perioperative period. In addition, there appeared to be no reversible element to this risk.

## COMMENT

In this article, we have presented a multifactorial risk index for patients undergoing noncardiac surgical procedures, which we have applied to patients in our general medical consultation service. Our study has demonstrated the predictive validity of this index.[4] Clinicians who wish to use this index should do so with the following warnings. We recommend that clinicians who apply this index to their patients should define the relevant patient population for whom this index may be useful and estimate the pretest probabilities of perioperative cardiac complications among those patients for the various surgical procedures. It may be that the pretest probabilities for our institution and clinical setting (ie, a general medical consultation service) are not applicable. We expect that our pretest probabilities will be most similar to those for general medical and cardiology consultation services in other large teaching hospitals. These pretest probabilities will not be appropriate for a series of consecutive patients undergoing surgery, most of whom will have no previous history, symptoms, or signs of cardiac disease. We believe that some of the early difficulties encountered in applying the original index of Goldman and coworkers arose from an inadequate consideration of the effect of variations in pretest probabilities on posttest risks. It would be misleading to conclude that the index performs poorly, if one merely compares the posttest probabilities displayed in the original article[2] with those in other institutions undergoing a different mix of surgical procedures.[8] It is extremely important to consider variations in pretest probabilities to make the best use of the index.

Just as sensitivity and specificity are "prevalence-independent properties" of a test (ie, do not change as the pretest probability changes from one population to another, at least theoretically), so too are the likelihood ratios prevalence independent. Thus, while each clinical setting will require estimation of pretest probabilities, the likelihood ratios associated with these scores should not change and may be applied directly using the nomogram in the Figure.

The reader should be aware that these pretest probabilities and likelihood ratios are based on a finite sample of patients and thus have confidence limits associated with them. The numbers quoted in this paper are the best estimates. We have presented confidence limits for the pretest probabilities. Confidence limits for the likelihood ratios are presented elsewhere.[4]

Lastly, we reiterate that the calculated posterior risk estimates should not be quoted blindly in the face of clinical information that is not adequately considered by the index. The multifactorial index is merely a model or starting point for consideration of cardiac risk, which is useful in that it allows the clinician to concentrate on some key variables. However, this does not excuse the clinician from performing the complete appropriate history and physical examination and from considering other variables in the assessment of risk. The index may be most useful for residents and junior consultants who are faced with an enormous number of clinical variables that could be considered in assessing cardiac risk.

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